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Experiment 2: To Containerize Application using Docker

Aim:

To containerize application using Docker This

experiment maps to LO4.

LO4: Containerize, orchestrate and deploy applications using Docker, Docker Compose, Kubernetes and cloud infrastructure tools.

Theory:

Steps to install docker:

i. Navigate to the official Docker website (docker.com) and download the Docker Desktop installer for your operating system (Windows).
ii. Locate the downloaded installer file on your computer and double-click it to run the setup wizard.
iii. Follow the on-screen prompts. When asked, ensure the option to Use WSL 2 instead of Hyper-V is checked (for optimal performance on Windows).
iv. Wait for the installation to finish, click "Close," and restart your computer if the installer prompts you to do so.
v. After restarting, search for "Docker Desktop" in your system's Start menu and launch the application. Accept the service agreement to start the Docker engine.
vi. Open your terminal or PowerShell and type docker --version and press Enter. If it returns a version number, Docker is successfully installed and ready to use.

Docker:

Docker is an open-source platform that allows you to package an application and all its required dependencies into a single, standardized unit for software development. This ensures that the application runs exactly the same way regardless of where it is deployed—whether on your personal laptop, a testing environment, or a live production server. Instead of dealing with the old problem of "it works on my machine but not yours," Docker isolates the software so it behaves predictably everywhere.

Docker Images: A Docker image is a lightweight, standalone, and immutable (read-only) executable package. In theory, it is a static file that contains the complete set of instructions and components necessary to run a specific application.

An image is built from multiple stacked layers, where each layer represents a specific instruction—such as installing a dependency, setting an environment variable, or copying

application code. It encompasses the application code itself, the runtime environment, system tools, system libraries, and default configuration settings. Because it is read-only, an image cannot be changed once it is built.

Docker Containers: A container is the active, runtime execution of a Docker image. It is a standardized, isolated operating environment where the application actually runs and processes data.

In theory, when an image is executed, the Docker engine adds a thin, writable layer on top of the static image layers. This writable layer allows the active container to modify files, generate logs, and maintain a temporary state while it is running. Containers isolate the running software from the host operating system and from other containers, which ensures the application performs uniformly and predictably regardless of the underlying infrastructure.

Implementation:

1. docker --version

```
C:\Users\15L>docker --version
Docker version 25.0.3, build 4debf41
```

2. docker search ubuntu

```
C:\Users\15L>docker search ubuntu
NAME                              DESCRIPTION                                     STARS     OFFICIAL
ubuntu                            Ubuntu is a Debian-based Linux operating sys…   17859     [OK]
ubuntu/squid                      Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx                      Nginx, a high-performance reverse proxy & we…   141
ubuntu/bind9                      BIND 9 is a very flexible, full-featured DNS…   120
ubuntu/kafka                      Apache Kafka, a distributed event streaming …   62
ubuntu/zookeeper                  ZooKeeper maintains configuration informatio…   15
ubuntu/apache2                    Apache, a secure & extensible open-source HT…   104
ubuntu/prometheus                 Prometheus is a systems and service monitori…   80
ubuntu/jre                        Chiseled Java runtime based on Ubuntu. Long-…   25
ubuntu/dotnet-aspnet              Chiselled Ubuntu runtime image for ASP.NET a…   26
ubuntu/python                     A chiselled Ubuntu rock with the Python runt…   33
ubuntu/mysql                      MySQL open source fast, stable, multi-thread…   74
ubuntu/postgres                   PostgreSQL is an open source object-relation…   45
ubuntu/redis                      Redis, an open source key-value store. Long-…   26
ubuntu/dotnet-runtime             Chiselled Ubuntu runtime image for .NET apps…   24
ubuntu/dotnet-deps                Chiselled Ubuntu for self-contained .NET & A…   16
ubuntu/grafana                    Grafana, a feature rich metrics dashboard & …   12
ubuntu/memcached                  Memcached, in-memory keyvalue store for smal…   7
ubuntu/jdk                        Ubuntu rock for OpenJDK                         0
ubuntu/rabbitmq                   Ubuntu-based RabbitMQ image - an open source…   0
ubuntu/mimir                      Ubuntu ROCK for Mimir, a horizontally scalab…   0
ubuntu/loki                       Grafana Loki, a log aggregation system like …   2
ubuntu/traefik                    Ubuntu ROCK for Traefik, a modern HTTP rever…   3
ubuntu/opentelemetry-collector    This is a rock for the OpenTelemetry Collect…   0
ubuntu/alertmanager               Ubuntu Rock for Alertmanager. It handles ale…   1
```

3. search ubuntu --filter stars=50

```
C:\Users\15L>docker search ubuntu --filter stars=50
NAME                DESCRIPTION                                     STARS     OFFICIAL
ubuntu              Ubuntu is a Debian-based Linux operating sys…   17859     [OK]
ubuntu/squid        Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx        Nginx, a high-performance reverse proxy & we…   141
ubuntu/bind9        BIND 9 is a very flexible, full-featured DNS…   120
ubuntu/kafka        Apache Kafka, a distributed event streaming …   62
ubuntu/apache2      Apache, a secure & extensible open-source HT…   104
ubuntu/prometheus   Prometheus is a systems and service monitori…   80
ubuntu/mysql        MySQL open source fast, stable, multi-thread…   74
```

4. docker search redis

```
C:\Users\15L>docker search redis
NAME                              DESCRIPTION                                     STARS     OFFICIAL
redis                             Redis is the world's fastest data platform f…   13604     [OK]
redis/redis-stack-server          redis-stack-server installs a Redis server w…   104
redis/redis-stack                 redis-stack installs a Redis server with add…   162
redis/redisinsight                Redis Insight - our best official GUI for Re…   54
redislabs/redis                   Clustered in-memory database engine compatib…   46
redis/rdi-operator                                                                0
redis/rdi-collector-initializer    Init container for RDI Collector               0
redis/rdi-api                                                                     0
redis/rdi-monitor                                                                 0
redis/rdi-collector-api                                                           0
redis/rdi-processor                                                               1
redis/rdi-cli                                                                     0
redis/rdi-flink-processor                                                         0
redis/reloader                                                                    0
redis/rdi-metrics-aggregator                                                      0
redis/rdi-flink-collector                                                         0
circleci/redis                    CircleCI images for Redis                       18
redis/sidekiq-benchmark           Sidekiq load benchmark in Rust - throughput …   0
redis/redis-benchmark-go          redis-benchmark utility in go.                  1
redis/vector-db-benchmark         Framework for benchmarking vector search eng…   1
bitnamicharts/redis               Bitnami Helm chart for Redis(R)                 37
redis/bcast-tracking-bench        Redis CLIENT TRACKING BCAST invalidation lis…   0
redis/redis-di-cli                                                                0
redis/pubsub-sub-bench                                                            0
cimg/redis                                                                        2
```

5. docker pull redis

```
C:\Users\15L>docker pull redis
Using default tag: latest
latest: Pulling from library/redis
062e450697fa: Pull complete
c1af3958d6f8: Pull complete
234dd6735104: Pull complete
6f0ec8fc933c: Pull complete
8fd80e87c6f1: Pull complete
4f4fb700ef54: Pull complete
fd6b3803006b: Pull complete
Digest: sha256:234c902a2db49461a129e2d4aeff85b28cf20187ed274a67f6e50995fa713c7b
Status: Downloaded newer image for redis:latest
docker.io/library/redis:latest
```

6. docker run -d --name my_database redis

```
C:\Users\15L>docker run -d --name my_database redis
9b0e3e996f293b18b60014144219f5584a96112d489a06eb9ad31ec7e07cbf85
```

7. docker ps

```
C:\Users\15L>docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED          STATUS          PORTS      NAMES
9b0e3e996f29   redis     "docker-entrypoint.s…"   40 seconds ago   Up 40 seconds   6379/tcp   my_database
```

8. docker inspect my_database

```
C:\Users\15L>docker inspect my_database
[
    {
        "Id": "9b0e3e996f293b18b60014144219f5584a96112d489a06eb9ad31ec7e07cbf85",
        "Created": "2026-07-21T05:03:18.0895840752",
        "Path": "docker-entrypoint.sh",
        "Args": [
            "redis-server"
        ],
        "State": {
            "Status": "running",
            "Running": true,
            "Paused": false,
            "Restarting": false,
            "OOMKilled": false,
            "Dead": false,
            "Pid": 890,
            "ExitCode": 0,
            "Error": "",
            "StartedAt": "2026-07-21T05:03:18.428780659Z",
            "FinishedAt": "0001-01-01T00:00:00Z"
        },
        "Image": "sha256:da3958f512eed039170b888ab48a80fa3424482ee50d52a634cf71c7b7b1c
        "ResolvConfPath": "/var/lib/docker/containers/9b0e3e996f293b18b60014144219f558
```

9. docker logs -f my_database

```
C:\Users\15L>docker logs -f my_database
Starting Redis Server
1:C 21 Jul 2026 05:03:18.446 * oO0OoO0OoO0Oo Redis is starting oO0OoO0OoO0Oo
1:C 21 Jul 2026 05:03:18.446 * Redis version=8.8.0, bits=64, commit=00000000, modified=1, pid=1, just started
1:C 21 Jul 2026 05:03:18.446 * Configuration loaded
1:M 21 Jul 2026 05:03:18.446 * monotonic clock: POSIX clock_gettime
1:M 21 Jul 2026 05:03:18.447 * Running mode=standalone, port=6379.
1:M 21 Jul 2026 05:03:18.447 * <bf> RedisBloom version 8.8.0 (Git=unknown)
1:M 21 Jul 2026 05:03:18.447 * <bf> Registering configuration options: [
1:M 21 Jul 2026 05:03:18.447 * <bf>      { bf-error-rate       :      0.01 }
1:M 21 Jul 2026 05:03:18.447 * <bf>      { bf-initial-size     :       100 }
1:M 21 Jul 2026 05:03:18.447 * <bf>      { bf-expansion-factor :         2 }
1:M 21 Jul 2026 05:03:18.447 * <bf>      { cf-bucket-size      :         2 }
1:M 21 Jul 2026 05:03:18.447 * <bf>      { cf-initial-size     :      1024 }
1:M 21 Jul 2026 05:03:18.447 * <bf>      { cf-max-iterations   :        20 }
1:M 21 Jul 2026 05:03:18.447 * <bf>      { cf-expansion-factor :         1 }
1:M 21 Jul 2026 05:03:18.447 * <bf>      { cf-max-expansions   :        32 }
1:M 21 Jul 2026 05:03:18.447 * <bf> ]
1:M 21 Jul 2026 05:03:18.447 * Module 'bf' loaded from /usr/local/lib/redis/modules//redisbloom.so
1:M 21 Jul 2026 05:03:18.449 * <search> search-workers default: 16 (min of MAX_WORKER_THREADS=16 and CPU cores)
1:M 21 Jul 2026 05:03:18.449 * <search> Redis version found by RedisSearch : 8.8.0 - oss
1:M 21 Jul 2026 05:03:18.449 * <search> RediSearch version 8.8.0 (Git=d2ae026)
1:M 21 Jul 2026 05:03:18.449 * <search> Low level api version 1 initialized successfully
1:M 21 Jul 2026 05:03:18.449 * <search> gc: ON, prefix min length: 2, min word length to stem: 4, prefix max expansions: 200,
 query timeout (ms): 500, timeout policy: return, oom policy: return, cursor read size: 1000, cursor max idle (ms): 300000, m
ax doctable size: 1000000, max number of search results:  1000000, default scorer: BM25STD,
1:M 21 Jul 2026 05:03:18.449 * <search> Initialized thread pools!
1:M 21 Jul 2026 05:03:18.449 * <search> Enabled workers threadpool of size 16
1:M 21 Jul 2026 05:03:18.453 * <search> Subscribe to config changes
```

10. docker exec -it my_database sh

```
C:\Users\15L>docker exec -it my_database sh
# exit
```

11. stop my_database

```
C:\Users\15L>docker stop my_database
my_database
```

12. docker ps --all

```
C:\Users\15L>docker ps --all
CONTAINER ID   IMAGE                     COMMAND                  CREATED         STATUS                     PORTS
               NAMES
9b0e3e996f29   redis                     "docker-entrypoint.s…"   4 minutes ago   Exited (0) 23 seconds ago
               my_database
8ddb1e79e05f   ubuntu                    "/bin/bash"              19 hours ago    Exited (0) 19 hours ago
               distracted_ishizaka
298abd2b1a9d   c234a24851ba              "docker-entrypoint.s…"   3 months ago    Exited (0) 3 months ago
               naughty_shirley
1a197c2bd0f5   c234a24851ba              "docker-entrypoint.s…"   3 months ago    Exited (255) 19 hours ago   6379/tc
p              happy_galois
013ed608bab8   c234a24851ba              "docker-entrypoint.s…"   4 months ago    Exited (0) 4 months ago
               nifty_yonath
f9b579837c27   d1397258b209              "docker-entrypoint.s…"   4 months ago    Exited (0) 4 months ago
               vigilant_knuth
06b8f6ece314   ca2b0f26964c              "/bin/bash"              2 years ago     Exited (255) 4 months ago
               happy_cerf
7b8921bf4fb0   d1397258b209              "docker-entrypoint.s…"   2 years ago     Exited (255) 3 months ago   6379/tc
p              jatinn
c782c1137761   node                      "docker-entrypoint.s…"   2 years ago     Exited (0) 2 years ago
               jatingupta
ca1d141bfabd   node                      "docker-entrypoint.s…"   2 years ago     Exited (0) 2 years ago
               reverent_williams
7638b03139a0   d1397258b209              "docker-entrypoint.s…"   2 years ago     Exited (0) 2 years ago
               jatin
d53dc9583439   d1397258b209              "docker-entrypoint.s…"   2 years ago     Exited (0) 2 years ago
               happy_engelbart
45c72eb69cd7   d1397258b209              "docker-entrypoint.s…"   2 years ago     Exited (0) 2 years ago
               keen_blackburn
b9b0d800f902   d1397258b209              "docker-entrypoint.s…"   2 years ago     Exited (0) 2 years ago
               priceless_jackson
a91dfde56008   d1397258b209              "docker-entrypoint.s…"   2 years ago     Exited (0) 2 years ago
```

13. docker rm my_database docker rmi redis

```
C:\Users\15L>docker rm my_database
my_database

C:\Users\15L>docker rmi redis
Untagged: redis:latest
Untagged: redis@sha256:234c902a2db49461a129e2d4aeff85b28cf20187ed274a67f6e50995fa713c7b
Deleted: sha256:da3958f512eed039170b888ab48a80fa3424482ee50d52a634cf71c7b7b1ca45
Deleted: sha256:8c7dd802be726678ecfba0631bd31608bd65c7deeb6dab90edb0f587b85b8d24
Deleted: sha256:2f094725331acf70ef1e0ddcaf7e37801d3cc785c26bd27595e76daf6c9fa713
Deleted: sha256:8992420a62a1fd65b4362a8707b4b3b50deff9f016e1c17dce1006b64ed498c2
Deleted: sha256:e53415315ef5fad7b12dc96ca27f8992e1cd17aa4a3bdc687119015633f4966a
Deleted: sha256:524505f4cf2205e269c812e3e4443f971e7245d7b97b9d09a7563419a6b18676
Deleted: sha256:ca8bda56b662384291dce284b44aba53d024597d1eef7dc4d9424af92ab93a03
Deleted: sha256:f2ec4de84f559f5c7be4233b589cdbdbb5507807e05621b77320edd55a1f2a0f
```

14. images

```
C:\Users\15L>docker images
REPOSITORY                  TAG                                       IMAGE ID       CREATED         SIZE
ubuntu                      latest                                    de7345b16e94   7 days ago      100MB
sample-flask-app            v1                                        f7d64ef4ebbb   3 months ago    141MB
myredis                     version1                                  5c08b3fad205   3 months ago    140MB
redis                       <none>                                    c234a24851ba   4 months ago    140MB
hello-world                 latest                                    1b44b5a3e06a   11 months ago   10.1kB
node                        latest                                    8bbb9d1bc633   2 years ago     1.1GB
nginx                       1.0                                       5682b6dc087e   2 years ago     169MB
ubuntu                      <none>                                    ca2b0f26964c   2 years ago     77.9MB
ubuntu                      <none>                                    3db8720ecbf5   2 years ago     77.9MB
mysql                       latest                                    019814493c7a   2 years ago     632MB
httpd                       latest                                    2776f4da9d55   2 years ago     167MB
redis                       <none>                                    d1397258b209   2 years ago     138MB
docker/welcome-to-docker    latest                                    c1f619b6477e   2 years ago     18.6MB
getting-started             latest                                    8d8a2af2207f   3 years ago     265MB
docker/getting-started      latest                                    3e4394f6b72f   3 years ago     47MB
docker/desktop-git-helper   5a4fca126aadcd3f6cc3a011aa991de982ae7000   efe2d67c403b   4 years ago     44.2MB
```

15. docker search nginx

```
C:\Users\15L>docker search nginx
NAME                                DESCRIPTION                                     STARS     OFFICIAL
nginx                               Official build of Nginx.                        21340     [OK]
nginx/nginx-ingress                 NGINX and  NGINX Plus Ingress Controllers fo…   122
nginx/nginx-prometheus-exporter     NGINX Prometheus Exporter for NGINX and NGIN…   52
nginx/nginx-ingress-operator        NGINX Ingress Operator for NGINX and NGINX P…   4
nginx/nginxaas-loadbalancer-kubernetes                                              1
bitnamicharts/nginx                 Bitnami Helm chart for NGINX Open Source        4
ubuntu/nginx                        Nginx, a high-performance reverse proxy & we…   141
kasmweb/nginx                       An Nginx image based off nginx:alpine and in…   9
rancher/nginx                                                                       4
linuxserver/nginx                   An Nginx container, brought to you by LinuxS…   236
dtagdevsec/nginx                    T-Pot Nginx                                     0
paketobuildpacks/nginx                                                              0
vmware/nginx                                                                        3
gluufederation/nginx                 A customized NGINX image containing a consu…   1
cleanstart/nginx                    Secure by Design, Built for Speed, Hardened …   0
activestate/nginx                   ActiveState's customizable, low-to-no vulner…   0
antrea/nginx                        Nginx server used for Antrea e2e testing        0
intel/nginx                                                                         0
docksal/nginx                       Nginx service image for Docksal                 1
geokrety/nginx                      Our customized nginx image                      0
circleci/nginx                      This image is for internal use                  2
ilios/nginx                         Nginx customized to run Ilios along with the…   0
corpusops/nginx                     https://github.com/corpusops/docker-images/     1
dockette/nginx                      Nginx SSL / HSTS / HTTP2                        3
wayofdev/nginx                      Nginx Docker image for PHP development. SSL-…   0
```

16. docker pull nginx

```
C:\Users\15L>docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
062e450697fa: Pull complete
82454cdbf456: Pull complete
3c7ab7949321: Pull complete
cacfcdd01f30: Pull complete
b6698f04e005: Pull complete
2bedaf25031a: Pull complete
d26f27cc8c41: Pull complete
Digest: sha256:5a88c9c45479443d7be2eadc894b4ed0a9801bae03d97a5760ae13b5c2005942
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
```

17. docker run -d --name my_webserver nginx

```
C:\Users\15L>docker run -d --name my_webserver nginx
05aa2b9e49e9ddff7e2dd2ab49bbad9adc23074311a3899d9b73037cdf7e86f4
```

18. exec -it my_webserver sh

```
C:\Users\15L>docker exec -it my_webserver sh
# exit
```

19. docker stop my_webserver docker rm my_webserver

```
C:\Users\15L>docker stop my_webserver
my_webserver

C:\Users\15L>docker rm my_webserver
my_webserver
```

20. docker rmi nginx docker images

```
C:\Users\15L>docker rmi nginx
Untagged: nginx:latest
Untagged: nginx@sha256:5a88c9c45479443d7be2eadc894b4ed0a9801bae03d97a5760ae13b5c2005942
Deleted: sha256:4e5db4761e0ff445f7fd29aad680ad28e8abf7d204895557f145d65535abcc1c
Deleted: sha256:2b3d10d90a77d4926b101279e7a65873c42b162087f614f75e88fa5075578458
Deleted: sha256:ce85d358b2fc8993b1f4258150e18ad68036164a2c1c5ec6cfe9f73b38c79038
Deleted: sha256:06f06fccf12160f5fb2d54c5810fee894e7171b9a3eb1dc07a9895af95f39fb4
Deleted: sha256:2d98d2a2e4c526409607174e7167b3961d4bd120ea1183772b248435b1d4ae7a
Deleted: sha256:4fad4b776a969a9a24fe73959a688c19525df696299519da94b5dd9224a74828
Deleted: sha256:135e056db1b5ffaad85c4e52f57983f61a937e246d79a5609cd5e042be60574a
Deleted: sha256:f2ec4de84f559f5c7be4233b589cdbdbb5507807e05621b77320edd55a1f2a0f

C:\Users\15L>docker images
REPOSITORY                TAG                                        IMAGE ID       CREATED         SIZE
ubuntu                    latest                                     de7345b16e94   7 days ago      100MB
sample-flask-app          v1                                         f7d64ef4ebbb   3 months ago    141MB
myredis                   version1                                   5c08b3fad205   3 months ago    140MB
redis                     <none>                                     c234a24851ba   4 months ago    140MB
hello-world               latest                                     1b44b5a3e06a   11 months ago   10.1kB
node                      latest                                     8bbb9d1bc633   2 years ago     1.1GB
nginx                     1.0                                        5682b6dc087e   2 years ago     169MB
ubuntu                    <none>                                     ca2b0f26964c   2 years ago     77.9MB
ubuntu                    <none>                                     3db8720ecbf5   2 years ago     77.9MB
mysql                     latest                                     019814493c7a   2 years ago     632MB
httpd                     latest                                     2776f4da9d55   2 years ago     167MB
redis                     <none>                                     d1397258b209   2 years ago     138MB
docker/welcome-to-docker  latest                                     c1f619b6477e   2 years ago     18.6MB
getting-started           latest                                     8d8a2af2207f   3 years ago     265MB
docker/getting-started    latest                                     3e4394f6b72f   3 years ago     47MB
docker/desktop-git-helper 5a4fca126aadcd3f6cc3a011aa991de982ae7000   efe2d67c403b   4 years ago     44.2MB
```

Conclusion:

In this experiment, the foundational concepts of Docker and application containerization were successfully studied and practically applied. The theoretical framework established the role of Docker in providing standardized, isolated environments, while clearly distinguishing between static, immutable Docker images and active, writable Docker containers.

Through the hands-on implementation phase, a complete container lifecycle was executed using a standard database image (redis,nginx). The practical steps successfully demonstrated how to:

Search for and pull specific images from a registry.

Deploy an image into an active, isolated container running in the background.

Monitor, inspect, and access the internal shell of a live container.

Perform proper environment cleanup by stopping the container, removing it, and deleting the underlying image.

Ultimately, this experiment provided the working knowledge necessary to package, deploy, and manage isolated software environments consistently across different systems.